

Brussels, 19 May 2025

COST 075/25

DECISION

Subject: Memorandum of Understanding for the implementation of the COST Action “Pan-European Network for Inflammaging: A Multi-omics Integration Approach” (INFLAMomx) CA24166

The COST Member Countries will find attached the Memorandum of Understanding for the COST Action Pan-European Network for Inflammaging: A Multi-omics Integration Approach approved by the Committee of Senior Officials through written procedure on 19 May 2025.

MEMORANDUM OF UNDERSTANDING

For the implementation of a COST Action designated as

COST Action CA24166

**PAN-EUROPEAN NETWORK FOR INFLAMMAGING: A MULTI-OMICS INTEGRATION APPROACH
(INFLAMomx)**

The COST Members through the present Memorandum of Understanding (MoU) wish to undertake joint activities of mutual interest and declare their common intention to participate in the COST Action, referred to above and described in the Technical Annex of this MoU.

The Action will be carried out in accordance with the set of COST Implementation Rules approved by the Committee of Senior Officials (CSO), or any document amending or replacing them.

The main aim and objective of the Action is to consolidate resources and knowledge on noncommunicable diseases (NCDs) by investigating inflammaging as a key shared mechanism, using advanced multi-omics integration approaches to enable a unified, cross-disciplinary understanding of disease initiation and progression. This will be achieved through the specific objectives detailed in the Technical Annex.

The present MoU enters into force on the date of the approval of the COST Action by the CSO.

OVERVIEW

Summary

The prevalence of noncommunicable diseases (NCDs) rises with age, placing a significant burden on public health. As the population ages, an increase in co- and multimorbidity is anticipated, making prevention a key scientific priority at the European level. In recent years, considerable attention has been given to inflammaging, a chronic, low-grade inflammation associated with aging, which contributes to the development of various NCDs, including cardiovascular diseases, neurodegenerative disorders, and metabolic conditions. The primary aim of INFLAMomx is to consolidate resources on NCDs through a Pan-European collaborative network and to investigate them as consequences of inflammaging using cutting-edge multi-omics approaches. This network will unite diverse European expertise across scientific fields (molecular biology, bioinformatics, and healthcare), while fostering collaboration across all career stages. Building upon European research initiatives, INFLAMomx will maximize the use of existing resources, strengthening computational biology and inflammaging research across Europe. The outcomes will enhance understanding of the molecular mechanisms driving inflammaging and its role in the development of NCDs. Furthermore, it will lay the foundation for innovations and translation into clinical practice. A key focus of INFLAMomx is the establishment of standardized guidelines for multi-omics integration methods and the training of the next generation of scientists with an integrative mindset, bridging knowledge gaps between wet lab researchers and computational scientists. Moreover, through targeted dissemination activities aimed at various stakeholders – general public, patient organizations, media, and policymakers – the project will raise awareness about the importance of maintaining health and pave the way for the development of preventive healthcare strategies.

Areas of Expertise Relevant for the Action	Keywords
• Biological sciences: Systems biology • Biological sciences: Bioinformatics • Basic medicine: Quantitative genetics • Basic medicine: Databases, data mining, data curation, computational modelling • Clinical medicine: Non-communicable diseases	• Inflammaging • Systems biology • Multi-omics biomarkers • Noncommunicable diseases • Personalized medicine

Specific Objectives

To achieve the main objective described in this MoU, the following specific objectives shall be accomplished:

Research Coordination

- To contribute to a deeper understanding of the inflammatory molecular mechanisms of aging which will provide insight into the possibilities to slow down or completely delay the onset of NCDs, which in the long run will lead to a relief of the health system.
- To establish an online repository of relevant multi-omics studies on inflammaging and relevant NCDs which will be used by the Action Participants and the whole scientific community.
- To identify multi-omics biomarkers of inflammaging and relevant NCDs.
- To optimize and standardize protocols and pipelines for multi-omics integration methods which will result in guidelines of best practices and reproducibility.
- To define knowledge gaps in the field of multi-omics integration and inflammaging and propose priority areas for implementation into the clinical setting.
- To promote, raise awareness and facilitate the adoption of INFLAMomx approach aiming for a paradigm

shift where chronic diseases would be evaluated as a manifestation of single underlying process of inflammaging, leading to a more unified and efficient strategies for prevention and treatment, easing their burden on the healthcare system.

Capacity Building

- To establish and expand a multidisciplinary expert network to facilitate knowledge exchange among clinicians, biologists, bioinformaticians, computer scientists, industry professionals, policy makers, professional societies, and educational non-profit organizations (e.g. student associations, patient association groups); to promote interdisciplinary thinking and achieve breakthroughs in the field of inflammaging.
- To perform knowledge transfer among experts from multidisciplinary fields to enhance collaboration between basic science and clinics, and between industry and academia, with the goal of understanding, improving, and developing protocols that address the role of inflammaging in NCDs.
- To develop an online platform for distributing knowledge, sharing data, exchanging experiences, and tracking the Action progress, with a focus on recruiting and providing career development opportunities for master's and PhD students, YRIs, and ITC.
- To serve as a crucial link between researchers and key stakeholders, including policy makers, industry, healthcare, and patient organizations, to advance understanding of inflammaging in disease initiation and progression, raise global awareness of its prevention, and build capacity for future collaborative and innovative research through publications, conferences and social media.

TECHNICAL ANNEX

1. S&T EXCELLENCE

1.1. SOUNDNESS OF THE CHALLENGE

1.1.1. DESCRIPTION OF THE STATE OF THE ART

Aging population as a global burden: Chronic diseases, also known as noncommunicable diseases (NCDs), are the leading causes of mortality and disability in developed countries. According to the World Health Organization (WHO), common chronic conditions such as cardiovascular diseases, cancer, chronic respiratory diseases, and diabetes account for approximately 80% of premature deaths. **Each year, 41 million people die from the consequences of NCDs, among whom 17 million people die before age 70. The economic burden of NCDs is vast and growing, with direct and indirect costs reaching hundreds of billions of euros in the EU annually¹.** Additionally, the prevalence of comorbidity or multimorbidity, where individuals suffer from two or more chronic conditions simultaneously, is rising². **Moreover, the risk for developing NCDs increases with aging.**

NCDs are often referred to as chronic or complex diseases because they typically progress slowly over time, are lifelong conditions, share common risk factors, require long-term management instead of a cure, and often occur simultaneously. NCDs emerge from a complex interplay of genetic, environmental, and social factors. Also, modern lifestyle behaviours such as smoking, unhealthy diets, physical inactivity, and excessive alcohol consumption significantly contribute to their development. These various environmental and biological factors influence NCDs initiation and progression with aging. **One of the emerging concepts of aging is inflammaging, which suggests that aging at the molecular level is driven by chronic low-grade inflammation, a consequence of the gradual decline in immune system function.** The main concept is that the inflammatory triggers behind inflammaging can come from external sources, such as chronic viral infections, diet, and lifestyle. However, the majority are likely produced internally, originating from self-debris generated by the continuous turnover of cells and tissues. This phenomenon is particularly important because all the major age-related diseases, such as cardiovascular conditions, neurodegenerative disorders, and cancers, have an inflammatory pathogenic component. Age is a major risk factor for these diseases, and since chronic low-grade inflammation is a key determinant to many of them, **it is evident that targeting inflammaging offers a strategic approach to combat age-related diseases collectively, rather than addressing them individually³.** At the molecular level, epigenetic modifications (e.g., DNA methylation), which are highly influenced by age, play a critical role in maintaining this inflammatory environment. People who are epigenetically older than their chronological age are more susceptible to age-related diseases⁴. Age-related changes in the microbiome also play a role, as disruptions in microbiota composition have been linked to immune dysfunction and diseases such as Alzheimer's^{5,6}. The theory of inflammaging is supported by studies of healthy centenarians, for which evidence shows that they are less susceptible to developing NCDs, exhibit optimal blood markers that suggest a lack of inflammation, have diverse gut microbiomes, and are epigenetically younger than their chronological age⁷.

Chronic diseases could emerge simultaneously, making it essential to examine shared molecular profiles to provide a broader understanding of their connections⁸. **However, studies investigating co-and multimorbidity as the main outcome are rare⁹.** Findings suggest that multimorbidity affecting the same physiological systems is more likely to share genetic variants, while those affecting different physiological systems tend to share genetic components at the network level. **Network medicine thus proposes that chronic diseases should be viewed as one common entity, regardless of their symptomatology (i.e. which organ is primarily affected by the disease), due to the proposed phenomena of shared genes network.** Therefore, from the aspect of inflammaging theory, it would be particularly important to investigate which biological networks are associated with the development of age-related multimorbidity.

Underexplored potential of omics data: High-throughput technologies and genome-wide association studies (GWAS) have made significant progress in understanding the genetic basis of various diseases. In the past two decades, more than 4,000 GWAS have been published, uncovering associations between over 50,000 genetic variants and numerous complex traits and diseases. However, despite the promise of these genomic discoveries, their clinical translation has been slower than anticipated. Consequently, research has increasingly shifted toward other omics, including **epigenomics, transcriptomics,**

proteomics, metabolomics, and metagenomics. Analyzing the combined contributions of different omics seems to be essential for a more holistic understanding of health and disease. **This integrative approach is at the core of systems biology**, which has led to the emergence of a new medical paradigm: **P4 medicine** (predictive, preventive, personalized, and participatory medicine)¹⁰. The adoption of multi-omics integration approaches has resulted in the development of a wide range of methods, tools, and platforms for analyzing, visualizing, and interpreting multi-omics data¹¹. These methods implement statistical, network-based, machine learning (ML) and artificial intelligence (AI) techniques on the integrated different omics layers to elucidate key multi-omics features associated with complex traits¹². Briefly, strategies for multi-omics integration can be grouped based on the dominant statistical method (concatenation, network/knowledge, joint dimension reduction, deep learning/AI), the anchor for integration (horizontal, vertical, diagonal, mosaic), or the timing (early, middle, late)^{11,13,14}. **However, despite the variety of methodologies available, there is still a lack of sufficient comparative analysis between them, which would in turn lead towards harmonization and standardization of protocols and automated pipelines.**

Therefore, to better understand the biological mechanisms behind inflammaging and its role in the development of NCDs, a multidisciplinary network must be established to drive breakthrough insights. Currently, the NCDs research community is fragmented, with expertise spread across clinicians, molecular biologists, and computational biologists, each focused on specific chronic diseases. **Also, there is immense potential stored in the generated omics data within different NCD research groups. To fully harness this potential, it is crucial to gather these resources and implement integrative bioinformatics algorithms in the field of inflammaging.**

1.1.2. DESCRIPTION OF THE CHALLENGE (MAIN AIM)

Current demographic trends predict that the average life expectancy in Europe will continue to rise, resulting in a growing proportion of elderly individuals within the population. This shift carries significant economic and healthcare implications. **Therefore, it is crucial to move from a reactive approach focused on curing NCDs to a proactive one that emphasizes understanding health and implementing effective prevention strategies.** The most appropriate approach to address these challenges is to form a multidisciplinary network that will **change the focus from a disease-centric view towards a health-centric view** (with prevention being the key), **and from a single disease view towards a co- and multimorbidity view** (with inflammaging as key shared mechanism).

To make this significant step forward, INFLAMomx will implement the most recent technological advances of multi-omics data integration methods with use case on inflammaging and several relevant NCDs. Despite the vast majority of available methodologies, multi-omics data integration poses a unique challenge, because a one-size-fits-all approach does not work. Several obstacles remain: it is still not fully known how different omics layers correlate within the same sample, different technologies result in varying numbers of features and statistical properties, each omics dataset has intrinsic technological limitations and noise structures, there are inconsistent data preprocessing standards, lack of guidelines or recommendations for practical application of multi-omics data integration, lack of multi-omics algorithms which can be translated into clinical setting, and lack of multi-omics “ground-truth” for assessing reliability. **This highlights the urgent need for standardized protocols in data processing, integration, and validation to ensure reliable, reproducible results.** Moreover, extracting and interpreting meaningful insights from these rich and diverse data sets is complex, requiring expertise across multiple disciplines.

Therefore, the main aim of the Action is to consolidate resources on NCDs through a pan-European collaborative network and to investigate them through the lens of inflammaging as one of the key drivers of disease initiation and progression, using cutting-edge multi-omics approaches. By uniting researchers from diverse fields – spanning NCDs, inflammation, aging, bioinformatics, systems biology, and computational sciences – INFLAMomx aims to integrate knowledge and stimulate innovative ideas for addressing these complex health challenges. Through various collaborative activities and knowledge exchange, the INFLAMomx aims to: 1) investigate concepts of inflammaging as a shared mechanism for many NCDs, 2) shift the paradigm from a disease-centric to a health-centric view, 3) establish standardized guidelines for multi-omics integration methods, 4) set the path for translating multi-omics biomarkers into the clinical setting, and 5) train the next generation of scientists with an integrative mindset.

1.2. PROGRESS BEYOND THE STATE OF THE ART

1.2.1. APPROACH TO THE CHALLENGE AND PROGRESS BEYOND THE STATE OF THE ART

INFLAMomx aims to build a top-level and unique network of both established and young researchers in the field of inflammaging, driven by the community's novelty and dynamism, as well as the opportunity to conduct competitive research in computational biology without requiring access to expensive facilities. It will attract investigators across diverse areas of expertise such as NCDs, inflammation, aging, bioinformatics, and systems biology, as well as industrial representatives and policymakers. The Action will create **an intellectual platform** assembling the expertise, methodologies, technologies, existing literature, omics data (e.g. genome, epigenome, transcriptome, microbiome, proteome), and bioinformatics protocols into the joint repository. By systematising these resources, the Action will facilitate the research on inflammaging within the network. **Particularly, a large amount of omics data related to inflammation and aging is already present in the publicly available databases as well as from the research cohorts of the Action Participants, however, without a communication platform, they will remain scattered and the added value from their merging will be lost.**

The multi-omics analysis of inflammaging and NCDs will lead to the main research outcome of the Action – understanding the key determinants of healthy aging and age-related diseases. A comprehensive understanding of inflammaging and related diseases could improve patients' wellbeing, disease treatment and prevention, and mitigate the healthcare economic burden. The results of the Action can in long run serve as a **backbone for preventive public health programs** at the regional and national levels, contribute to the translation of primary knowledge to the clinical setting to **preserve health and delay the onset of age-related diseases**, and inform the relevant experts for more appropriate adoption of future health policies. The Action will create **novel hypotheses and identify molecules** that could be further studied as **potential disease biomarkers and drug targets**, triggering further efforts towards their investigation in clinical trials. Earlier research has shown that people who are biologically older than their chronological age are more susceptible to the development of various diseases associated with aging, but also that such epigenetic changes are reversible and can be used for the development of interventions that will slow aging⁴. **The reversibility of the molecular modifications induced by aging and identification of relevant pathways associated with age-related NCDs could open the avenue for developing innovative solutions for healthy senescence and disease prevention.**

INFLAMomx aims to drive innovation and improvement into the field of multi-omics data analysis in NCDs. Gaining knowledge from multi-omics, which share unique features like high dimensionality, complexity, and heterogeneity, is a significant challenge that needs protocol standardization. Therefore, the Action **will focus on the generation and dissemination of recommendations, guidelines, suggestions for standard operating procedures (SOPs), and best-practice pipelines that enable the integration of different molecular profiles into a common data network**, which is necessary for a complete understanding of the molecular mechanisms of inflammaging. Moreover, establishing a new research network that deals with bioinformatics and statistical processing of omics data will **strengthen the field of computational and systems biology** across Europe. **Established tenured professors will mentor less experienced Young Researchers and Innovators (YRIs)**, guiding them toward cutting-edge bioinformatics, multi-omics integration, and systems biology methods to help bridge knowledge gaps, gain valuable experience, and facilitate their mobility into environments crucial for their future career development in computational and systems biology.

The Action will foster collaborative work on manuscripts intended to be published in high-impact journals. Working Group (WG) leaders will include less experienced researchers and those from Inclusiveness Target Countries (ITC) in the manuscript preparation, thus **increasing their international visibility**. YRIs involved in the Action implementation **will develop a wide network with national and international researchers** and research groups in the field of inflammation and aging, systems biology, and bioinformatics. Most of all, **the obtained results, international contacts, and experience gained during the implementation of the Action will open the path for applying for new competitive EU and international programmes, ensuring the long-term financial sustainability of the newly established research collaborations.**

The Action will organise **trainings and short-term scientific mobilities** for research staff involved in the implementation. These trainings are strategically designed not only to support the Action's immediate goals and deepen expertise within the research groups, but also to **foster long-term collaboration beyond their completion**. The Action will host workshops that bring together members to **address key challenges critical to achieving the Action goals**, such as establishing and validating multi-omics protocols, standardizing best practices for sample collection, and optimizing data generation. Additionally, **open discussion sessions** will be organized between experts, policymakers, and patients to **raise awareness about inflammaging**, share cutting-edge knowledge on the biological mechanisms of health and disease, and present the latest omics technologies and tools addressing these issues. Dissemination efforts will also prioritize **educating young researchers** by collaborating with student organizations and **engaging the general public** through annual science popularization events.

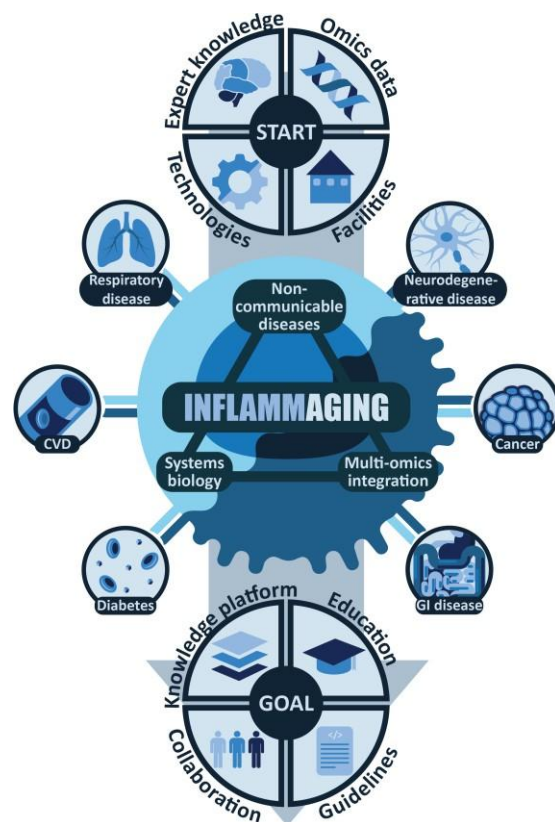


Figure 1. The INFLAMomx network pathway to impact

Taken together, INFLAMomx will not only offer new insights from the multi-omics approach in inflammaging, but also create a structured environment that supports European researchers studying non-communicable and age-related diseases. It will help them advance their research in established or novel directions and empower them to lead this cutting-edge field into the future.

1.2.2. OBJECTIVES

1.2.2.1. Research Coordination Objectives

INFLAMomx aims to create a Pan-European network of multidisciplinary expert working groups to establish standardized bioinformatic guidelines for reproducible research in the multi-omics setting, with a specific focus on inflammaging and its role in NCDs initiation and progression. The Action will initially focus on three major NCDs as a pilot study, and which established resources will later be used to include other NCDs. Specific research coordination objectives are:

- **RCO1: To contribute to a deeper understanding of the inflammatory molecular mechanisms of aging** which will provide insight into the possibilities to slow down or completely delay the onset of NCDs, which in the long run will lead to a relief of the health system.
- **RCO2: To establish an online repository of relevant multi-omics studies on inflammaging and relevant NCDs** which will be used by the Action Participants and the whole scientific community.
- **RCO3: To identify multi-omics biomarkers** of inflammaging and relevant NCDs.
- **RCO4: To optimize and standardize protocols and pipelines for multi-omics integration methods** which will result in guidelines of best practices and reproducibility.
- **RCO5: To define knowledge gaps in the field of multi-omics integration and inflammaging** and propose priority areas for implementation into the clinical setting.

- **RCO6: To promote, raise awareness and facilitate the adoption of INFLAMomx approach aiming for a paradigm shift** where chronic diseases would be evaluated as a manifestation of a single underlying process of inflammaging, leading to a more unified and efficient strategies for prevention and treatment, easing their burden on the healthcare system.

1.2.2.2. Capacity-building Objectives

The capacity-building aims of the Action are:

- **CBO1: To establish and expand a multidisciplinary expert network** to facilitate knowledge exchange among clinicians, biologists, bioinformaticians, computer scientists, industry professionals, policy makers, professional societies, and educational non-profit organizations (e.g. student associations, patient association groups); to promote interdisciplinary thinking and achieve breakthroughs in the field of inflammaging.
- **CBO2: To perform knowledge transfer** among experts from multidisciplinary fields to enhance collaboration between basic science and clinics, and between industry and academia, with the goal of understanding, improving, and developing protocols that address the role of inflammaging in NCDs.
- **CBO3: To develop an online platform** for distributing knowledge, sharing data, exchanging experiences, and tracking the Action progress, with a focus on recruiting and providing career development opportunities for master's and PhD students, YRIs, and ITC.
- **CBO4: To serve as a crucial link** between researchers and key stakeholders, including policy makers, industry, healthcare, and patient organizations, to advance understanding of inflammaging in disease initiation and progression, raise global awareness of its prevention, and build capacity for future collaborative and innovative research through publications, conferences and social media.

2. NETWORKING EXCELLENCE

2.1. ADDED VALUE OF NETWORKING IN S&T EXCELLENCE

2.1.1. ADDED VALUE IN RELATION TO EXISTING EFFORTS AT EUROPEAN AND/OR INTERNATIONAL LEVEL

INFLAMomx will strive to gather all available information on inflammaging from the existing efforts at the European and international levels to maximize the use of European resources. Numerous initiatives under Horizon Europe and previous Horizon 2020 framework programs are related to exploring aging, however, these consortia were mostly focusing on specific diseases like Alzheimer's or had a broader scope on healthy aging. **A smaller number of EU projects are addressing inflammaging specifically, and these specific projects are of particular interest to the Action:** "MARK-AGE - European Study to Establish Biomarkers of Human Ageing", "PROPAG-AGEING The continuum between healthy ageing and idiopathic Parkinson Disease within a propagation perspective of inflammation and damage", "RESETageing RESEArch for healThy AGEING", "Gut-InflammAge Age-associated signatures in the composition and pro-inflammatory status of the gut microbiome in humans and mice, and the impact of a periodic fasting intervention to promote healthy aging", "Let T Be Letting up senescence and inflammaging through T cells", "MyTissue Studying the metabolism and function of myeloid dendritic cells and neutrophils in distinct tissues in health and aging". The Action will connect to these previously funded and ongoing initiatives, to consolidate **findings from these disease-specific projects and investigate inflammaging as a common mechanism underlying multiple NCDs.** Also, establishing positive synergies with existing European initiatives is a major objective of the Action which will be accomplished through knowledge dissemination, invitations to join Working Groups, and proposals for joint efforts where appropriate.

INFLAMomx will also actively reach out to other complementary COST Actions and invite their participants to join and engage in the Action activities and events. Previously funded and ongoing COST Actions complementary to the Action, whose members will be approached, are: CA23119 SENESENCE2030 (related to senescence in preventing age-related diseases), CA20117 Mye, InfoBank, and CA21147 ENOTTA (related to inflammation biomarkers and inflammatory chronic

disease), while of particular interest are CA21165 PerMediK and CA21153 AtheroNET, both of which rely on the implementation of a multi-omics approach in chronic kidney disease and atherosclerotic cardiovascular disease, respectively. In comparison to the existing efforts, **the added value of INFLAMomx lies in the systematisation of knowledge related to multi-omics approaches in studying NCDs not as a disease-specific phenomenon but as a cross-cutting shared mechanism of inflammaging that affects many aspects of aging.**

INFLAMomx members are also extensively involved in previous and ongoing large-scale projects and longitudinal biobanks, which will also be one of the fundamental pillars in achieving the Action objectives. In addition, some of the interesting data-related European initiatives that the Action will seek collaborations with are: European Chronic Disease Alliance, European Institute for Systems Biology and Medicine, 1+Million Genomes (1+MG) initiative, EATRIS – European infrastructure for translational medicine, BBMRI-ERIC as a European research infrastructure for biobanking, ELIXIR as a European life sciences infrastructure, NCD Risk Factor Collaboration (NCD-RisC), and others which will be identified during the Action.

The Action will focus on **complementary technological expertise and knowledge transfer** available in non-ITC and ITC. By **promoting inclusiveness, excellence, and uniqueness** of each Action Participant, INFLAMomx will have a significant added value on the networking in the field of inflammaging and multi-omics integration. By integrating additional collaborations through various ongoing European and international initiatives, INFLAMomx **participants will additionally increase their visibility and further develop new and reliable networks.** By focusing on more advanced integrative multi-omics analyses and systemic perspectives, INFLAMomx will **fill critical knowledge gaps.** It will integrate the data to uncover previously undetected patterns in inflammaging, which in turn will provide a **holistic understanding of aging as a systemic process driven by inflammation that affects different organs in interconnected ways.** These efforts could significantly improve the understanding of how inflammation drives aging and NCDs, ultimately **leading to novel preventive or therapeutic strategies.** By empowering interdisciplinary collaboration and collaboration with industry, INFLAMomx could **enhance the translation of multi-omics biomarkers into market-ready interventions.**

2.2. ADDED VALUE OF NETWORKING IN IMPACT

2.2.1. SECURING THE CRITICAL MASS, EXPERTISE AND GEOGRAPHICAL BALANCE WITHIN THE COST MEMBERS AND BEYOND

INFLAMomx formed a large, top-level, robust and interdisciplinary network of research institutions, business enterprises (small- or medium-sized; SMEs), non-profit organisations and governmental organisations. The Action consists of senior researchers who are experienced PIs and will provide a key role as expert advisors with policymaking potential, mid-career researchers as younger PIs with excellent track record, and YRIs who will participate in all of the Action activities, which will deepen their knowledge, research and management skills. This multidisciplinary network comprises of:

- **Molecular biologists** with expertise in specific NCDs, inflammation, and/or aging (including both wet lab and dry lab researchers);
- **Bioinformaticians and computational biologists** working in different omics areas, spanning from the application of available omics methods and tools to developing new multi-omics algorithms;
- **Systems biologists** working with advanced generative modelling and using AI approaches in the field of multi-omics integration;
- **Clinicians and healthcare researchers** with expertise in applying a multi-omics approach to translate the findings from basic medicine into clinical settings;
- **Entrepreneurs** involved in the development of multi-omics tools and bioinformatic pipelines, providing various types of IT services, software development, and cloud computing;
- **Members of governmental organisations** involved in improving health economics, developing preventive strategies, and translating personalized medicine into clinical practice.

In addition to the expertise needed for successful implementation of the Action, several large cross-sectional and prospective cohorts are available among the Action Participants (comprising subjects with NCDs, subjects with various co- and multimorbidities, healthy subjects), with their clinical data and multi-omics profiles. The involvement of existing research infrastructures in the Action will enhance the broader dissemination and exploitation of the Action outcomes (knowledge hub, guidelines of best practices) by hosting them in established curated tools, ensuring transparency, reliability, and maximizing the impact of the Action results.

INFLAMomx network primarily **bridges the gap between wet and dry lab researchers** in the common task of explaining the molecular mechanisms underlying various NCDs, with a special focus on multi-omics integration methodology.

The Action will continue to seek actively and will welcome additional relevant participants affiliated as COST Full or Cooperating Members, Near Neighbour Countries, and International Partner Countries, which will provide additional value to the whole network. **Continuous efforts will be made to involve YRIs, ensuring a balanced gender representation throughout the Action's duration.**

2.2.2. INVOLVEMENT OF STAKEHOLDERS

A wide variety of stakeholders interested in the Action aims and objectives will be actively involved and continuously complemented, while collaborative interaction between them will be facilitated. In order to ensure the stakeholder engagement aligns with the implementation of the Action objectives and to maximize its impact on scientific and societal levels, a broad range of stakeholders will be involved through stakeholder involvement plans and brokerage events:

- **Researchers at academic institutions or research institutions** will provide scientific expertise in inflammaging or bioinformatics, contribute to knowledge exchange, training of YRIs, and dissemination. They will be invited to participate in the Action meetings, Workshops, Training Schools, and Conferences.
- **Clinicians and health sector**, in particular those involved in NCDs and age-related diseases, will be central to promote and disseminate the long-term Action aims to shift focus from disease-centric to health-centric view, and to encourage out-of-the-box way of thinking that low-grade chronic inflammation might be one of the key targetable elements of many NCDs. They will be approached through conferences, various professional societies, and will be invited to participate in the Action Meetings, Conferences, Workshops, and dissemination events.
- **Industry** (SME and other companies) that develop software for various omics analysis tools and bioinformatic pipelines, but also biopharmaceutical companies which can support knowledge transfer and real-world application of research outcomes. Technology transfer events and hackathons will be used to attract a wider industry sector.
- **Young Researchers and Innovators (YRIs)** will continuously be monitored, and new members will be invited to join the Action, will be encouraged to take responsible and leading roles and participate in decision-making processes. Career development opportunities through Training Schools and STSMs will offer them interdisciplinary training, expertise, and better positioning on the future job market. They will participate in all Action activities.
- **Funders and Policy Makers** whose participation will help ensure that the Action aims and deliverables contribute to evidence-based policymaking and that they align with regulatory frameworks. They will actively participate in advisory roles and will be invited to the Action Conferences and Workshops, during which designated round table discussions will be held. Representatives from health, research, and education organizations and ministries at the regional, national, European, and international level will be invited. Engagement with international funding agencies, the European Commission, and global institutions such as the World Bank will ensure the Action is aligned with global strategies and agendas, enabling access to additional funding opportunities and international collaboration.
- **Non-governmental organizations (NGOs), scientific societies and patient associations** will participate in dissemination activities and promote the Action's aims and objectives. They will be invited to participate in the Action Conferences and Workshops, during which designated round table discussions will be held. They will also be involved in raising public awareness,

outreach activities, public engagement events, and capacity-building initiatives.

- **General Public, including Media**, represents key stakeholders, whose engagement is essential in the processes of data gathering, sharing, and management, as it builds trust, ensures ethical data practices, and empowers patients to actively shape how their data is used for research and healthcare advancements. Raising awareness of the novel approach provided within the Action, together with the innovation potential and translational capabilities, can have a large benefit on the well-being of citizens within Europe and globally. They will be invited to participate in public outreach events organized by the Action, as well as already established initiatives at the national and European level (e.g. European research night, science festivals).

To promote active involvement of all stakeholders, annual conferences, dissemination, and outreach events will be held to share the Action progress, disseminate the results of various activities and to engage with a wider audience. The Action online platform will be used to facilitate continuous communication and collaboration among all stakeholders, to provide access to key Action documents and to serve as a repository for Action deliverables. The Action social media will be used to further reach and interest broader audiences.

To ensure sustainability after the completion of the Action, strategies for long-term collaboration will be developed, e.g. formal agreements will be established, continuation of knowledge transfer and dissemination will be encouraged and promoted, and collaborative follow-up projects and joint proposals for future funding opportunities will be pursued.

3. IMPACT

3.1. IMPACT TO SCIENCE, SOCIETY AND COMPETITIVENESS, AND POTENTIAL FOR INNOVATION/BREAKTHROUGHS

3.1.1. SCIENTIFIC, TECHNOLOGICAL, AND/OR SOCIOECONOMIC IMPACTS (INCLUDING POTENTIAL INNOVATIONS AND/OR BREAKTHROUGHS)

Scientific impact: The approach and results of the Action will contribute to a **paradigm shift** from *classifying and treating diseases related to aging and inflammation as separate entities* to the *identification and exploitation of common molecular signatures* of these diseases. This approach will serve as a basis for understanding the mechanisms of healthy aging, disease initiation and progression, and the emergence of co- and multimorbidity (living with two or more chronic illnesses). The main **methodological asset** of the Action is the **integration of all molecular profiles into a joint network**. This will result in the identification of common biomarkers and targets for **prevention, diagnosis and personalised treatment strategies** applicable to different NCDs, which may not share the same symptomatology. Building on this approach, one of the major scientific impacts of the Action is the identification of novel and the systematisation of known **biological pathways related to inflammaging by using a multi-omics integration approach**. Identification of disease-causing factors within molecular networks will foster entrepreneurship, driving both **innovative diagnostics solutions and development of targeted therapeutics**. By establishing a curated, high-quality online repository of multi-omics studies on inflammaging, healthy aging, and related NCDs, INFLAMomx will have a transformative impact on AI applications in NCDs. This comprehensive dataset will enable more accurate predictive models, enhance personalised treatment strategies, and accelerate the discovery of novel biomarkers and therapeutic targets in the NCD field.

Technological impact: The action will produce an **online platform** available for the Action Participants and for outside experts and non-experts. The platform will consist of two main collections: 1) a collection of **relevant multi-omics studies and data on inflammaging, healthy aging, and relevant NCDs**, 2) a collection of **protocols for integrative data analysis, data storage and data sharing** as well as names and affiliation of researchers who contributed to these activities. In particular, the protocols section will include bioinformatic scripts for data analysis, with benchmarking efforts encouraged to provide validated pipelines. Best practices protocols related to data storage and sharing will ensure the protection of the personal data of human subjects, information on storage details, and efficient data sharing among the Action Participants. Sharing unpublished data for joint integrative analysis will help extract useful insights from the data and deepen collaboration among the Action Participants. In general, the **communication platform will be open to all stakeholders (researchers, clinicians, patient organizations, industry and policy makers) interested in inflammaging to find resources,**

collaborators and solutions to their problems. In particular, the platform will foster match-making between wet lab and dry lab researchers, and data analysts and bioinformaticians having complementary expertise, as well as collaboration between the private sector and academia. **Startups and spinout companies** focused on integrative data analysis are expected to emerge. More importantly, greater impact is expected from biotech companies leveraging the knowledge of key players of the inflammaging molecular network, providing innovative diagnostics solutions and targeted therapies. Overall, the platform will result in **better usage of human resources and other capacities within and outside the network, with a great potential to outlive the 4-year life of the Action.**

Socioeconomic impact: Prevention of diseases related to inflammaging is at the centre of this action. Vast knowledge emerging from molecular biology studies on different NCDs is not sufficiently integrated to fully understand the key drivers of inflammaging. The results of the Action will bring up molecular biomarkers associated with healthy aging that could be translated into clinical settings. With **machine learning based models** that incorporate socioeconomic and lifestyle characteristics along with patients' molecular profile data, improved identification of high-risk groups will be enabled. Through numerous dissemination and communication activities (such as Researcher's Night, social networks, podcasts and conventional media), it will contribute to **increasing awareness** of the importance of preserving one's health and the possibilities of **preventing chronic diseases** among the general public. In the long term, this can affect the reduction of the frequency of chronic diseases and the need to use specialized health care. This will improve the quality of life among the European increasingly aging population and reduce healthcare costs. Also, it will provide a healthy late career stage workforce, which can increase productivity and alleviate pension systems across Europe.

The expected impact of the Action aligns with the priorities of the EU for Health Civil Society Alliance and the United Nations' Sustainable Development Goals, particularly Goal 3: "Ensure healthy lives and promote well-being for all at all ages." The Action will adhere to the principles of P4 medicine by focusing on personalized approaches to patient care and raising awareness about the importance of healthy lifestyles and disease prevention. The Action aims to **create a resource for healthcare innovation, with bioinformatic methods facilitating the translation of basic research into clinical applications.**

3.2. MEASURES TO MAXIMISE IMPACT

3.2.1. KNOWLEDGE CREATION, TRANSFER OF KNOWLEDGE AND CAREER DEVELOPMENT

The Action will leverage all available training tools – meetings, workshops, training schools, STSMs and conference grants – as outlined in the guidelines, to build a comprehensive multidisciplinary network of research groups and participants from diverse countries and backgrounds, fostering knowledge creation, exchange, and new career opportunities for YRIs and those from ITC.

Knowledge creation: The international and multidisciplinary Management Committee (MC), Working Groups (WGs), and dedicated YRIs will collaborate closely to explore the role of inflammaging in the development of NCDs and co- and multimorbidity. **Annual in-person and online MC and WG meetings will drive knowledge creation**, with progress and outputs publicly available on the Action's online platform, along with reports and opportunities for new members and stakeholders to get involved. The Action will consist of five Working Groups, of which WG1 will focus on knowledge hub creation around relevant multi-omics studies and data on inflammaging, WG2 will create optimised and standardized protocols for best practices in multi-omics integration methodology to enhance reproducibility, WG3 will use the outputs created in WG1 and WG2 to identify the key multi-omics biomarkers for inflammaging which clinical potential will be discussed, WG4 will work closely with WG2 in order to create educational materials and develop training schools to foster knowledge transfer of best methodological practices to YRIs, whereas WG5 will serve as a platform for communication, dissemination, and outreach.

Transfer of knowledge: The Action will ensure efficient **knowledge transfer among stakeholders by facilitating the development of a multi-omics knowledge hub and best practices through Workshops and Training Schools.** The YRI committee will play a pivotal role in promoting networking and advancing cutting-edge skills for young scientists from ITCs, particularly by involving them in protocol design and the presentation of key findings during conferences and training schools. To maximize the knowledge transfer, members will organize **three educational Workshops** and **six**

Training Schools, especially designed for YRI scientists (master's students, PhDs, and postdocs), covering key topics aligned with the Action aims. All activities will adhere to FAIR principles (Findability, Accessibility, Interoperability, and Reuse) to ensure transparency and reproducibility of developed protocols and shared datasets through the Action online platform. Young researchers from ITCs will be encouraged and supported through **at least two STSMs per grant period** and given opportunities to present their findings and newly acquired skills at international conferences. Furthermore, **at least four Conference Grants per grant period** will be awarded to promote the inclusion of YRIs from ITCs.

Career development: The Action will strengthen Europe's research competitiveness by fostering the career development of its Participants, with a focus on young researchers. YRI will be offered leadership roles to enhance their management skills and gain exposure to an interdisciplinary, international environment. Through COST-supported training programs in areas such as medicine, bioinformatics, and mathematical modelling, and short visits between academia and biotech companies, researchers will acquire advanced technical and entrepreneurial skills, boosting their career prospects. The Action will also promote personalized mentoring by connecting YRIs with senior experts from academia and industry, facilitating collaborations, cross-regional projects, and job opportunities. With strong engagement from industry partners and startup experts, early-stage members will be encouraged to pursue business initiatives in molecular healthcare. This international, interdisciplinary network will be ideally positioned to apply for Horizon Europe and other major research programs.

3.2.2. PLAN FOR DISSEMINATION AND/OR EXPLOITATION AND DIALOGUE WITH THE GENERAL PUBLIC OR POLICY

Measures to maximize impact will be divided into 1) **activities aimed at promoting the Action and raising awareness** to a wider audience, including the general public and media, and 2) activities aimed at **raising interest among various stakeholders and exploitation-oriented dissemination** provided by the innovative approaches and methodologies.

The scientific breakthroughs and activities of the Action will be made publicly available through several different approaches:

- Providing an online platform that offers an overview of all the Action activities, actively pursued research questions, scientific results and contact information to participating research institutions.
- Disseminating results to a wider interested public via the Action online platform and social networks (LinkedIn, X formerly known as Twitter).
- Promotional brochures, infographics and a booklet for various stakeholders (in particular for non-specialists) will be developed.
- Striving for publications in open-access and high-rated (Q1) internationally recognized journals in the field of genetics, aging, bioinformatics and personalized medicine.
- Presenting results obtained during the Action implementation at international conferences such as the Conference of the European Society of Human Genetics (ESHG), international conferences led by the organizations FEBS or EMBO/EMBL (in the field of systems biology, aging, inflammation, etc.), International/European Conference on Computational Biology (ISMB/ECCB), Belgrade Bioinformatics Conference (BelBi), NGSymposium for young researchers or similar conferences in Europe.
- Involving public authorities and policymakers to introduce the core concepts of preventive and personalized medicine, to inspire strategic improvements in healthcare.
- Educating the general public about the biology of healthy aging through participation in popular science events such as the European Researchers' Night, Science Festival, open door day and visiting schools, etc.
- Educating students on inflammaging, systems biology, and bioinformatics by engaging in student science fairs and delivering lectures in master's and PhD programs.

Based on all the aforementioned dissemination activities, the international scientific community and the general public will have several different opportunities to get to know the Action expertise and research

results directly. Dissemination activities aimed at the general public are extremely important for longitudinal cohort studies, given that they rely on the voluntary participation of respondents who must always have a clearly defined interest and benefit from participating in such projects.

4. IMPLEMENTATION

4.1. COHERENCE AND EFFECTIVENESS OF THE WORK PLAN

4.1.1. DESCRIPTION OF WORKING GROUPS, TASKS AND ACTIVITIES

INFLAMomx will be structured in accordance with COST rules. In addition to mandatory leadership positions (i.e. Chair, Vice Chair, Grant Holder Scientific Representative, Management Committee, Working Group Leaders, Grant Awarding Coordinator, and Science Communication Coordinator), it will also include the Core Group (CG) which will oversee the collaborative activities between different Working Groups, ensuring timely implementation. Additional roles, e.g. Working Group Vice Leaders, will also be appointed. The Action will consist of 5 Working Groups (WG), of which WG1-3 will focus on Research Coordination Objectives, while WG4-5 will focus on Capacity-building Objectives. In particular, WG4 will be in charge of realization of training activities, and WG5 of communication and dissemination activities. Depending on the main objective of each WG, they will be led by an expert in inflammaging, bioinformatics, or translational research, respectively. Initial research focus will be on three major NCDs (as agreed upon with the Action Participants), which will serve as a pilot program. Information on multi-omics data of additional NCDs will be continuously added during and after the Action's completion, which will ensure long-term impact.

WG1: Knowledge hub creation

Main objective: To compile available information on relevant inflammaging multi-omics studies.

Tasks: **T1.1:** Define specific NCDs which incidence increases with aging as these diseases are of central interest for inflammaging research; **T1.2:** Identify and gather information from relevant multi-omics studies on inflammaging and NCDs defined in T1.1 (summary level data will be collected from public databases, from the Action Participants and other researchers will be contacted to promote sharing of their published and unpublished data); **T1.3:** Explore and identify molecular profiles and biological pathways associated with inflammaging; **T1.4:** Explore and identify patient characteristics associated with inflammaging; **T1.5:** Select the most relevant molecular markers of inflammaging; **T1.6:** Identify knowledge gaps and priority areas in the field of inflammaging; **T1.7:** Organize a conference “Inflammaging – past, present and future” (in collaboration with WG5).

Activities: **A1.1:** Review of literature, public databases, completed and ongoing EU projects and initiatives, and the Action Participants available data sources with regards to molecular profiles and clinical data associated with inflammaging and NCDs defined in T1.1; **A1.2:** Creation of harmonized INFLAMomx knowledge hub, in line with FAIR principles, from relevant multi-omics studies on inflammaging and NCDs defined in T1.1; **A1.3:** Write review paper on the state-of-the-art and knowledge gaps in the field of inflammaging; **A1.4:** Continuously monitor the scientific literature and conference dissemination materials with established bioinformatic pipeline for automated data mining of relevant datasets.

Milestones: **M1.1:** Data from the public databases gathered for the INFLAMomx knowledge hub; **M1.2:** Annual data update for the INFLAMomx knowledge hub.

WG2: Integrative framework of data management and reproducibility enhancement

Main objective: To identify the best practices for multi-omics data integration, ensuring standardization and reproducibility.

Tasks: **T2.1:** Identify, compare and evaluate currently used softwares and methods for multi-omics integration; **T2.2:** Identify and evaluate the best practices for data management and reproducibility; **T2.3:** Optimize protocols and pipelines for multi-omics integration methods; **T2.4:** Standardize the use of multi-omics integration methods; **T2.5:** Define reporting standards when applying multi-omics data integration

methods; **T2.6:** Test reproducibility of multi-omics integration methods based on identified benchmark dataset from D1.1; **T2.7:** Identify knowledge gaps and priority areas in the field of multi-omics data integration; **T2.8:** Organize a conference “Systems biology symposium – from modelling to application” (in collaboration with WG5).

Activities: **A2.1:** Test and optimize methods for multi-omics data integration and their respective parameters; **A2.2:** Write guidelines of best practices for optimal usage of multi-omics data integration methods; **A2.3:** Write guidelines of best practices for data management and reproducibility; **A2.4:** Write review paper on the state-of-the-art and knowledge gaps in the field of multi-omics data integration; **A2.5:** Continuously monitor the scientific literature, software releases and conference dissemination materials with established bioinformatic pipeline for automated data mining of relevant tools.

Milestones: **M2.1:** Optimized protocols and pipelines for multi-omics integration methods.

WG3: Exploration of implementation and translational potential

Main objective: To set a path towards a successful implementation of multi-omics biomarkers into clinical settings.

Tasks: **T3.1:** Foster communication between bioinformaticians and clinicians or wet lab scientists to improve understanding of each other's requirements and potentials; **T3.2:** Explore available standardization protocols of multi-omics datasets in terms of samples (sample collection, biobanking, storage, sharing), laboratory protocols (sample processing, DNA/RNA/protein isolation, sequencing), data management with special focus on FAIR principles; **T3.3:** Based on the outcome of T3.2 and D1.1 create guidelines of best practices for the future inflammaging multi-omics studies; **T3.4:** Apply optimized protocols for multi-omics data integration from WG2 to (re-)analyse the existing multi-omics datasets identified in D1.1 and/or through datasets from Action Participants (in collaboration with WG2); **T3.5:** Apply meta-analytic approach, if applicable, to select the most relevant multi-omics markers of inflammaging and NCDs defined in T1.1 (in collaboration with WG2); **T3.6:** Based on the outcome of T3.4 and T3.5 assess the interindividual variability in various patients characteristics to start the personalized approach towards the understanding of health and disease; **T3.7:** Draft preliminary documentation of needs, expertise and risks for future clinical trials based on the outcome of T3.4 and T3.5; **T3.8:** Explore the potential of various bioinformatics tools to enhance the real-life use in clinical settings, e.g. modelling cause-effect relationship with Mendelian randomization, creation of polygenic and multi-omics risk scores (in collaboration with WG2); **T3.9:** Identify challenges and obstacles of translating and implementing the potential omic or multi-omics biomarker into clinical use; **T3.10:** Organize a final conference “Molecular mechanisms of aging – lessons learned from *in silico* models” (in collaboration with WG5).

Activities: **A3.1:** Review of available online data sources with regards to standardization protocols of multi-omics datasets; **A3.2:** Analyse identified multi-omics datasets, either from publicly available databases or through the Action Participants datasets, to identify multi-omics biomarkers of inflammaging and NCDs defined in T1.1; **A3.3:** Implement additional bioinformatic approaches to enhance the understanding of inflammaging and NCDs defined in T1.1 (meta-analysis, Mendelian randomization, polygenic and multi-omics risk scores); **A3.4:** Write guidelines of best practices for future inflammaging multi-omics studies; **A3.5:** Write review paper on the state-of-the-art, with special focus on challenges and obstacles, in the field of implementing the potential omic or multi-omics biomarker into clinical use.

Milestones: **M3.1:** Collection of available standardization protocols of multi-omics datasets; **M3.2:** Creation of a repository of multi-omics biomarkers of inflammaging and NCDs defined in T1.1.

WG4: Knowledge exchange, capacity building and training

Main objective: To provide training in the key INFLAMomx areas – multi-omics datasets in inflammaging, multi-omics integration methods, data management and reproducibility.

Tasks: **T4.1:** Define topics for Workshops and Training Schools; **T4.2:** Organize Workshops for the

Action Participants; **T4.3:** Organize three-day Training Schools for the Action Participants and non-members, with special focus on involvement of YRIs from ITCs and gender balance; **T4.4:** Organize one-week interdisciplinary Training Schools especially targeted for YRIs only; **T4.5:** Organize a hackathon on multi-omics data integration challenge.

Activities: **A4.1:** Conduct a survey among the Action Participants to assess the need and expertise for specific topics for Workshops and Training Schools (webinars and in-person meetings); **A4.2:** Design “grand challenge” topics for interactive and collaborative Workshops where brainstorming and discussion will be emphasized; **A4.3:** Design educational materials and define example datasets to be used for Training Schools; **A4.4:** Define a challenge, select multi-omics dataset and write documentation for hackathon.

Milestones: **M4.1:** Three Workshops; **M4.2:** Six Training Schools; **M4.3:** One hackathon.

WG5: Communication, dissemination and outreach

Main objective: To coordinate communication, dissemination and outreach activities to facilitate collaboration within the Action Participants and to maximize the Action's visibility among different non-members stakeholders.

Tasks: **T5.1:** Create and maintain the Action online dissemination channels for internal (the Action Participants) and external (all other relevant stakeholders) use – online platform and social media accounts (LinkedIn, X formerly known as Twitter); **T5.2:** Use online dissemination channels to advertise events, trainings, open calls for STSMs and to promote publications as the Action results; **T5.3:** Create and distribute biannual newsletters, leaflets, and infographics (adjusted for each stakeholders group) to promote the Action progress; **T5.4:** Compile information of expertise, sample data and accessible high-throughput technology of each Action Participant; **T5.5:** Collect information of relevant experts in the field of inflammation and/or systems biology who can provide additional value to the network; **T5.6:** Coordinate annual Action Meetings; **T5.7:** Coordinate the organisation of STSMs; **T5.8:** Coordinate presentations at conferences organised by third parties; **T5.9:** Organize monthly one-hour Webinars where the Action Participants (more experienced YRIs and senior experts) will present their career path and research focus in relation to the Action topic; **T5.10:** Organize annual Action Conference in collaboration with WG1, WG2 and WG3; **T5.11:** Coordinate the organisation of Workshops and Training Schools in collaboration with WG4; **T5.12:** Identify relevant funding opportunities (public and private funding calls); **T5.13:** Identify relevant conferences, society meetings, and other events for promoting systems biology approach to study inflammaging within the scientific community; **T5.14:** Identify of social events and science popularization events for promoting inflammaging to the general public; **T5.15:** Establish contact with representatives of the relevant policy makers, industry, NGOs and patients' organizations to facilitate the Action aims and maximize impact.

Activities: **A5.1:** Develop and implement communication, dissemination and outreach strategy; **A5.2:** Create “Who-is-who” brochure of the Action Participants; **A5.3:** Contact identified relevant stakeholders in the field of inflammation and/or systems biology, and ask them to join the Action; **A5.4:** Create reporting template summarizing key findings relevant for the Action objectives after attending conferences organised by third parties; **A5.5:** Create reporting template summarizing key output after STSM completion; **A5.6:** Collect all relevant materials from other WGs which should be disseminated via online dissemination channels or as a newsletter; **A5.7:** Regularly inform the Action Participants of relevant events and funding opportunities; **A5.8:** Create promotional booklet for general public to educate and raise awareness of inflammaging concepts and to promote healthy aging.

Milestones: **M5.1:** Online platform and social media accounts constructed and running; **M5.2:** Communication, dissemination and outreach strategy created.

4.1.2. DESCRIPTION OF DELIVERABLES AND TIMEFRAME

Deliverable	Description	WG	Month
D5.1	Online platform and social media accounts	WG5	4
D5.2	“Who-is-who” brochure of the Action Participants	WG5	7

D3.1	Collection of identified available standardization protocols of multi-omics datasets	WG3	12
D1.1	Report from Conference “Inflammaging – past, present and future” (in collaboration with WG5)	WG1	15
D1.2	Public release of INFLAMomx knowledge hub	WG1	18
D2.1	Publication on state-of-the-art and knowledge gaps in the field of multi-omics data integration	WG2	18
D3.2	Publication on the state-of-the-art advancements in translating omic and multi-omics biomarkers into clinical practice	WG3	24
D2.2	Report from Conference “Systems biology symposium – from modelling to application” (in collaboration with WG5)	WG2	27
D5.3	Promotional booklet for the lay public	WG5	27
D1.3	Publication on state-of-the-art and knowledge gaps in the field of inflammaging	WG1	30
D2.3	Publication of guidelines for best practices for optimal usage of multi-omics data integration methods	WG2	30
D3.3	Publication of guidelines for best practices for future inflammaging multi-omics studies	WG3	36
D4.10	Reports from the hackathon	WG4	39
D2.4	Publication of guidelines for best practices for data management and reproducibility	WG2	42
D3.4	Collection of identified multi-omics biomarkers of inflammaging and NCDs defined in T1.1	WG3	42
D3.5	Preliminary documentation for future clinical trials based on D3.4	WG3	45
D3.6	Report from final Conference “Molecular mechanisms of aging – lessons learned from <i>in silico</i> models” (in collaboration with WG5)	WG3	48
D4.1-3	Reports from the Workshops	WG4	9, 21, 33
D4.4-9	Reports from the Training Schools with a collection of all educational materials	WG4	12, 18, 24, 30, 36, 42
D5.4-7	INFLAMox Annual Reports	WG5	12, 24, 36, 48

4.1.3. RISK ANALYSIS AND CONTINGENCY PLANS

Risk description	Risk level	Impact	Contingency plan
Low engagement and commitment toward achieving the Action goals	Medium	High	Organize (online and in-person) Work Group meetings and Workshops more frequently to identify problems, allocate responsibilities and develop actionable plans and timelines.
The Action Participants not motivated to share their (unpublished) multi-omics data within the network for joint, integrative analysis	Medium	Medium	1) Identify reasons for reluctance to share omics data; 2) Provide clear guidelines for data protection and terms of data sharing (A2.2, A2.3); 3) Encourage data sharing by providing state-of-the-art data analysis by the network. If these measures fail, publicly available data sets can be used.
The Action fails to attract novel participants from underrepresented sectors (private sector, clinics, GO, NGO).	Medium	Medium	1) Advertise action goals on the action online platform, professional social media and conferences targeting underrepresented sectors; 2) Encourage the Action Participants to invite their collaborators and colleagues from the non-academic sector to join the Action.

Expertise deficiency related to multi-omics data analysis among the Action Participants	Low	High	1) Intensify efforts to attract European and other researchers and innovators interested in multi-omics to join the action (A5.3); 2) Intensify trainings (through STSMs, training schools and workshops) to improve know-how among the current members.
Regional and gender balance disturbed	Low	Low	Provide more networking opportunities and leading positions to participants from underrepresented categories.
Low participation of YRIs	Low	Low	1) Through dissemination activities targeting YRIs, promote the Action goals, networking and career development opportunities; 2) Encourage COST Participants to invite their younger colleagues to join.

4.1.4. GANTT DIAGRAM

Activity	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Action meetings																
<i>Kick-off meeting</i>	KO															
<i>Annual meetings</i>					A				A				A			
<i>Final meeting</i>																F
<i>MC meetings</i>	MC		MC		MC		MC		MC		MC		MC			MC
<i>WG meetings</i>	WG		WG		WG		WG		WG		WG		WG			WG
<i>CG meetings</i>	CG	CG*	CG	CG*	CG	CG*	CG	CG*	CG	CG*	CG	CG*	CG	CG*	CG*	CG
<i>Calls for STSMs</i>			x		x		x		x		x		x			
WP1: Knowledge hub creation																
<i>WP1 Deliverables</i>					D1.1	D1.2				D1.3						
WP2: Integrative framework of data management and reproducibility enhancement																
<i>WP2 Deliverables</i>						D2.1			D2.2	D2.3				D2.4		
WP3: Exploration of implementation and translational potential																
<i>WP3 Deliverables</i>				D3.1				D3.2				D3.3		D3.4	D3.5	D3.6
WP4: Knowledge exchange, capacity building and training																
<i>WP4 Deliverables</i>			D4.1	D4.4		D4.5	D4.2	D4.6		D4.7	D4.3	D4.8	D4.10	D4.9		
WP5: Communication, dissemination and outreach																
<i>WP5 Deliverables</i>		D5.1	D5.2	D5.4				D5.5	D5.3			D5.6				D5.7

* online meetings